Feature Extraction
Local Descriptors

1. GOAL

- Introduction to local image descriptors. Descriptors based on image derivatives (edges,ridges).
- Introduction to descriptors based on convolution: Gabor Filters, Difference of Gaussians, Sobel

2. MATERIALS

Slides Intro_LocalDescriptors.ppt, python code (PyCode_LocalDescriptors.zip), which contains the main script Script_ImageDescriptors.py and functions for the computation and visualization of Gabor filter banks

3. SESSION EXERCISES

Exercise 1. Edges. Compute the gradient of the use images labelled 'filled_square', 'SAROI' without any filtering (filter_image =‘none’) and compare the results to the gradient obtained using a gaussian of sig=2, 4, 8 and a median filter of Medsze=3,5,7.

Exercise 2. Ridges and Valleys. Compute the Laplacian, Ridges (negative Laplacian) and Valleys (positive Laplacian) of the use images labelled 'filled_square', its opposite (1-'filled_square') and ‘square’. Compare results across use cases.

Exercise 3. Gabor Filters. Compute the default Gabor filter bank and:

a) Visualize the two banks of filters.

b) Apply the two filter banks to Compute the Laplacian, Ridges (negative Laplacian) and Valleys (positive Laplacian) of the use images labelled 'filled_square', its opposite (1-'filled_square') and ‘square’.

c) Visualize responses to each filter for the two Gabor filter banks and compare results to the ones obtained in Ex1 and Ex2. Would you use any of the Gabor filters to detect edges, valleys or ridges.

d) Repeat a)-c) using the alternative Gabor filter bank. What is the difference with the default Gabor filter bank? Would you use any of the Gabor filters to detect edges, valleys or ridges.

Exercise 4. Feature Spaces. Compute Otsu thresholding for the use image ‘SAROI’ and visualize the binarization. Analyse the histogram of SAROI intensity showing the lesion values in red and discuss if there exists a threshold (vertical line) able to perfectly separate lesion from other structures.

Consider for each pixel the pair of values given by SAROI intensity (im) and its Laplacian (Lap). Visualize the point cloud defined by assigning for each pixel a x-coordinate given by its intensity and a y-coordinate given by its Laplacian. Try to divide the plane with a line splitting (discriminating/classifying) the red (lesion) and blue points (background). Do you think you could achieve a more accurate separation than using only one feature (intensity)?